Arista Access Node WFD-ACC-48 QoS Congestion Event (Microbursts) Resolution SOP

Node: WFD-ACC-48 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_6792

Description

In the context of Arista's eOS, the Access Node WFD-ACC-48, situated within the Whitefield POP, is susceptible to QoS congestion events, including microbursts and interface tail drops, due to the role's inherent reliance on Quality of Service (QoS) mechanisms to manage traffic flow. As a critical component of the network infrastructure, this node's Access role necessitates the enforcement of strict QoS policies to ensure optimal throughput and minimize congestion-related disruptions. The occurrence of microbursts and interface tail drops can be attributed to the unpredictable nature of network traffic patterns, which, when left unmanaged, can lead to QoS queue starvation and subsequent congestion events. The specific alarm code ALARM_6792, indicative of a warning condition, signifies the potential for future congestion-related issues, necessitating prompt resolution to maintain network stability.

Pre-requisite Checks

- Verify the site-specific power and environmental conditions at Whitefield POP are within acceptable parameters.

- Confirm the redundant link status and ensure proper configuration to facilitate failover scenarios.

- Dispatch remote hands to Whitefield POP for on-site assistance, if required.

Potential Root Causes

Cause	Probability
QoS policy misconfiguration	High
Interface congestion due to excessive traffic	Medium
Hardware resource limitation (e.g., insufficient buffers)	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface WFD-ACC-48 | include packets, Expected Output: packets input: [x] packets output: [y], Description: This command aids in isolating interface-specific congestion issues by providing a quantitative assessment of packet flow, facilitating the identification of potential bottlenecks.

Command: show qos | detail, Expected Output: QoS policy configuration with detailed settings, Description: This command allows for the examination of the QoS policy configuration, facilitating the identification of potential misconfigurations or policy-related issues contributing to congestion events.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos | detail, Description: Review the QoS policy configuration to identify potential misconfigurations or policy-related issues contributing to congestion events.

Step 2: Step: Adjust QoS policy settings, Command: set qos policy , Description: Modify the QoS policy settings as necessary to address congestion-related issues, ensuring that policy revisions align with optimal traffic flow and throughput requirements.

Step 3: Step: Verify interface configuration, Command: show interface WFD-ACC-48 | include packets, Description: Confirm that interface settings are optimized for congestion-free operation, and adjust settings as necessary to address any identified interface-specific issues.

Post-Check Verification

```
Command: show qos | detail, Expected Output: QoS policy configuration with optimized settings, Description: Verify that QoS policy settings have been successfully updated and are now optimized for congestion-free operation.

Command: show interface WFD-ACC-48 | include packets, Expected Output: packets input: [x] packets output: [y] (indicating optimal interface operation), Description: Confirm that interface settings have been successfully updated and are now optimized for congestion-free operation.
```

Roll-back Procedures

- Step: Revert QoS policy configuration to previous settings, Command: set qos policy , Description: Revert QoS policy settings to the previous configuration, ensuring that any changes made during the resolution process do not adversely impact network operation.

- Step: Revert interface configuration to previous settings, Command: set interface WFD-ACC-48 , Description: Revert interface settings to the previous configuration, ensuring that any changes made during the resolution process do not adversely impact network operation.